WORKED SOLUTION

STEP III 1997, Question 13

Probability and Statistics • Normal Moment Generating Functions

STEP III 1997, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q13

The key idea

Everything rests on one integral: the moment generating function of a standard normal, found by completing the square in the exponent. After that, independence turns the mgf of a sum into a product, and the uniqueness of mgfs lets us read off distributions. The final part is the same calculation in disguise: a power of $Z = \mathrm{e}^{\mu+\sigma X}$ is just an exponential of a multiple of X.

Solution

Part (i)

Write $M(\theta) = \mathrm{E}(\mathrm{e}^{\theta X})$. Then

$$M(\theta) = \int_{-\infty}^{\infty} \mathrm{e}^{\theta x} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}x^2}\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(x^2-2\theta x)}\,\mathrm{d}x.$$

Completing the square, $x^2 - 2\theta x = (x-\theta)^2 - \theta^2$, so

$$M(\theta) = \mathrm{e}^{\frac{1}{2}\theta^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}(x-\theta)^2}\,\mathrm{d}x = \mathrm{e}^{\frac{1}{2}\theta^2} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}u^2}\,\mathrm{d}u,$$

substituting $u = x - \theta$. The last integral is the total integral of the standard normal density, which is 1. Hence, for every real θ,

$$\boxed{\mathrm{E}(\mathrm{e}^{\theta X}) = \mathrm{e}^{\frac{1}{2}\theta^2}.}$$

Part (ii)

Since X and Y are independent, so are $\mathrm{e}^{\theta a X}$ and $\mathrm{e}^{\theta b Y}$, and the expectation of a product of independent random variables is the product of their expectations. Using part (i) with θ replaced by $a\theta$ and by $b\theta$ (and the fact that Y has the same distribution as X),

$$\mathrm{E}(\mathrm{e}^{\theta(aX+bY)}) = \mathrm{E}(\mathrm{e}^{a\theta X})\,\mathrm{E}(\mathrm{e}^{b\theta Y}) = \mathrm{e}^{\frac{1}{2}a^2\theta^2}\,\mathrm{e}^{\frac{1}{2}b^2\theta^2} = \mathrm{e}^{\frac{1}{2}(a^2+b^2)\theta^2}.$$

Ask yourself Why does matching the mgfs settle the question of distribution?

A moment generating function that exists for all θ near 0 determines the distribution uniquely. So $aX + bY$ has the same distribution as X exactly when its mgf equals $\mathrm{e}^{\frac{1}{2}\theta^2}$ for all θ, that is, when

$$\mathrm{e}^{\frac{1}{2}(a^2+b^2)\theta^2} = \mathrm{e}^{\frac{1}{2}\theta^2} \quad \text{for all } \theta.$$

Taking logarithms and choosing any $\theta \neq 0$, this holds if and only if

$$\boxed{a^2 + b^2 = 1.}$$

(The "only if" direction is also clear without uniqueness: $\mathrm{Var}(aX + bY) = a^2 + b^2$ must equal $\mathrm{Var}(X) = 1$.) In general $aX + bY$ is normal with mean 0 and variance $a^2 + b^2$; for instance $(X + Y)/\sqrt{2}$ is again standard normal.

Part (iii)

For any real θ,

$$Z^\theta = \left(\mathrm{e}^{\mu+\sigma X}\right)^\theta = \mathrm{e}^{\mu\theta}\,\mathrm{e}^{(\sigma\theta)X},$$

and $\mathrm{e}^{\mu\theta}$ is a constant, so by part (i) with θ replaced by $\sigma\theta$,

$$\mathrm{E}(Z^\theta) = \mathrm{e}^{\mu\theta}\,\mathrm{E}(\mathrm{e}^{(\sigma\theta)X}) = \mathrm{e}^{\mu\theta}\,\mathrm{e}^{\frac{1}{2}\sigma^2\theta^2} = \mathrm{e}^{\mu\theta+\frac{1}{2}\sigma^2\theta^2},$$

as required.

Taking $\theta = 1$ gives the mean:

$$\boxed{\mathrm{E}(Z) = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2}.}$$

Taking $\theta = 2$ gives $\mathrm{E}(Z^2) = \mathrm{e}^{2\mu+2\sigma^2}$, so

$$\begin{aligned}\mathrm{Var}(Z) &= \mathrm{E}(Z^2) - (\mathrm{E}(Z))^2 = \mathrm{e}^{2\mu+2\sigma^2} - \mathrm{e}^{2\mu+\sigma^2} \\ &= \mathrm{e}^{2\mu+\sigma^2}(\mathrm{e}^{\sigma^2} - 1).\end{aligned}$$

Checks

- Differentiating $M(\theta) = \mathrm{e}^{\frac{1}{2}\theta^2}$ gives $M'(0) = 0$ and $M''(0) = 1$: mean 0, variance 1, as they should be.
- If $\sigma = 0$ then $Z = \mathrm{e}^{\mu}$ is constant; our formulas give mean e^{μ} and variance 0.
- Numerically, with $\mu = 0.3$ and $\sigma = 0.5$, integrating against the normal density gives $\mathrm{E}(Z) \approx 1.5296$ and $\mathrm{Var}(Z) \approx 0.6645$, matching $\mathrm{e}^{0.425}$ and $\mathrm{e}^{0.85}(\mathrm{e}^{0.25} - 1)$. A simulation of $0.6X + 0.8Y$ gave sample variance 1.000.

Common slips

- Writing $\mathrm{E}(Z) = \mathrm{e}^{\mu}$ by "plugging in the mean of X". In general $\mathrm{E}(g(X)) \neq g(\mathrm{E}(X))$, and here the convexity of the exponential pushes the mean up by the factor $\mathrm{e}^{\frac{1}{2}\sigma^2}$.
- Giving the condition as $a + b = 1$: it is the variances, not the coefficients, that add.

Beyond STEP

The random variable Z has the *log-normal* distribution, the standard model for quantities that grow by multiplying many small random factors, such as share prices in the Black–Scholes model. Part (ii) is a glimpse of a striking fact: the normal is essentially the only distribution with finite variance whose suitably scaled sums stay in the same family, which is tied up with the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, covers moment generating functions, the normal distribution and the central limit theorem at exactly this level.

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